

PROJECT ABSTRACT [Form - 1] SESSION (2026-27)

PROJECT ID: SKIT/CSE/2023-2027/...5.3... BRANCH: Computer Science & Engineering SECTION: ...D

TITLE OF PROJECT: Smart Pothole Detection SystemSDG MAPPING: Sustainable Cities and Communities

PROJECT TRACK:

R&D INNOVATION ☒	CONSULTANCY PROJECTS	DEPARTMENT AS CLIENT PROJECTS	CLIENT PROJECT	HARDWARE-BASED PROJECTS
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MANDATORY EXTERNAL EVALUATION OF PROJECT:

RESEARCH PAPER PUBLICATION ☒	PATENT FILING / GRANT	POSTER COMPETITION ☒
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PROBLEM STATEMENT:

To Develop an automated system that uses - vehicle-mounted camera and a deep learning model to detect potholes, tag their GPS location, store the results in a database, and generate road-condition score for each stretch - reducing dependence on manual surveys and citizen reporting.

TECHNOLOGY STACK:

NAME OF TOOL / TECHNOLOGY	FRONTEND/ BACKEND	SOFTWARE / HARDWARE	PURPOSE OF USE
Python	Backend	software	Core programming language for the entire pipeline
OpenCv	Backend	software	Extracts frames from road video for processing
YOLOv8	Backend	Software	AI model for detecting potholes in video frames
SQL	Backend	Software	Stores geotagged pothole detection records
Streamlit	Frontend	Software	web dashboard for video upload and result visualization.
Folium	Frontend	Software	Interactive map visualization of geotagged detections
Laptop (min. 8GB RAM)	-	Hardware	Runs model training and inference pipeline.
SmartPhone	-	Hardware	Records road video with synced GPS Track.

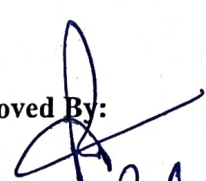
PROPOSED PROJECT SPRINTS:

S.No.	SPRINT NAME	PROPOSED USER STORY OF RESPECTIVE SPRINT
0	Data Collection & Model Training	To collect and label a pothole image dataset and fine-tune a YOLOv8 model, so that the system can accurately detect potholes.
1	Video Processing pipeline	To extract frames from road video, run detection on each frame, and log results into a database, so that raw video becomes structured, storable data.
2	Geotagging and Road scoring	To match each detection with its GPS location and compute a road-condition score per segment, so that road quality can be judged objectively.
3	Dashboard and Integration	To build a dashboard with video upload and an interactive map visualization, so that results can be reviewed visually.
4	Testing and Bug Fixing	To test the complete system end-to-end on real road footage and fix identified bugs, so that the final deliverable is reliable.

TEAM MEMBER DETAILS:

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Sapna Kumari	9079096606	Machine Learning, Computer vision, Python	AI ML Engineer & System Integrator dataset collection, YOLOv8 model training, video pipeline, road-condition scoring, map integration.
Sanyam Singh	9898959332	Data Annotation, Backend Development, web Dashboards	Data Engineer & Dashboard Developer dataset labeling, model evaluation, database design, GPS-matching logic, streamlit dashboard.

Verified & Approved By:


Pawan Kumar Patidar
Mentor's Name & Signature


18/8/26

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form - 2]

PROJECT ID: SKIT/CSE/2023-2027/...S3...BRANCH: Computer Science & Engineering SECTION: D

TITLE OF PROJECT: Smart Pothole Detection System

TEAM LEAD <u>Sapna Kumari</u>		SPRINT(s) NAME: Data Collection & model Training, Video Processing Pipeline, Geotagging and Road Scoring, Dashboard Development, Testing and Bug Fixing	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
To Collect & label Pothole dataset; begin model Training	10 Aug 2026	30 Oct 2026	Collected Pothole dataset from Kaggle & Roboflow; began fine-tuning the YOLOv8 model on the labeled data.
To Extract video frames & run detection	4 Oct 2026	31 Oct 2026	Built the frame extraction pipeline using OpenCV; ran YOLOv8 for inference on extracted frames
To Compute road-condition Score per stretch	1 Nov 2026	2 Jan 2027	Built the road-condition scoring module (Pothole Count, Size, Confidence) -> Score
To show detected potholes on map	3 Jan 2027	6 Feb 2027	Integrated Folium map visualization into the dashboard; plotted Geotagged detections
To test system end-to-end	7 Feb 2027	6 Mar 2027	Tested detection & scoring pipeline end-to-end on real road footage; fixed bugs found during testing
MEMBER1 <u>Sanyam Singh</u>		SPRINT(s) NAME: Data Collection & model Training, Video Processing Pipeline, Geotagging & Road Scoring, Dashboard Development, Testing and Bug Fixing	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
To Label dataset; evaluate model accuracy	10 Aug 2026	30 Oct 2026	Labeled/annotated Pothole images (Roboflow); evaluated the trained model's accuracy on test data
To Design database to store detections	4 Oct 2026	31 Oct 2026	Designed and implemented the SQL database schema; wired up detection logging into the database
To match each detection to GPS location	1 Nov 2026	2 Jan 2027	Implemented GPS-time stamp matching logic between detections and location track
To Build web interface to upload video & view results	3 Jan 2027	6 Feb 2027	Built the Streamlit dashboard (video upload, results display, layout)
To test dashboard end-to-end	7 Feb 2027	6 Mar 2027	Tested dashboard functionality end-to-end on real footage; fixed bugs found during testing.

Verified & Approved By:

Mentor's Name & Signature

Raman Kumar Patidar

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